Plastic pollution in Shanghai

Section 2.1

Khrishi Parmar– 21027654

Task 2.1 - orbital mechanics 3
Introduction 3
Region of interest 3
Payload 4
Orbital Trade-Off 6
Ground Station Location 8
Field of View 8
References 9

Task 2.1 - orbital mechanics

Introduction

This task began with a region of interest being selected that would be most appropriate for the mission objective chosen. Proceeding this research was conducted for different imaging tools and parameters decided upon to allow for the best outcome. An iterative process was used to define the optimal orbital elements to ensure that the coverage time was maximized over the chosen ground station and region of interest thus determining the overall mission design.

Region of interest

In 2019 Chinese state oceanic Administration declared 81% of China's coastal regions are heavily polluted with plastic debris (Wang, He and Sen, 2019). Shanghai specifically, became the city of interest as it lies on the Yangtze River Estuary and houses 2 main rivers that connect thousands of branching rivers in Shanghai (Luo et al., 2019). The Yangtze River Estuary channel plays an increasingly important role in national shipping and means waters are being polluted by ships' domestic sewage, garbage and other effluents (Zhang *et al.*, 2022).

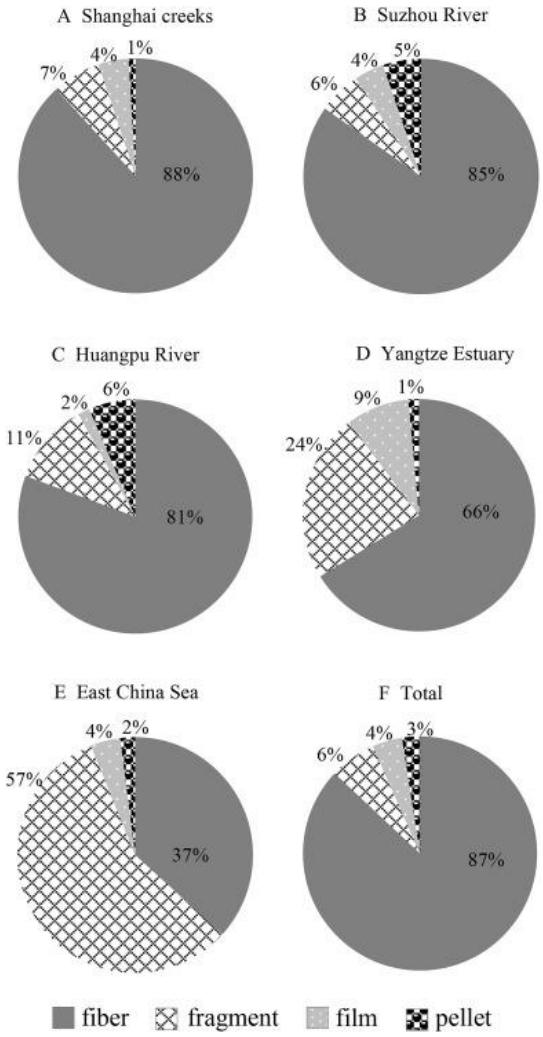


Figure 1: *depicts the different microplastic shapes found in each body of water (Luo et al., 2019).*

Figure 1 demonstrates how fiber was the most common material found in the bodies of water in Shanghai with an exception in the East China Sea. The different types of plastics found in the above marine areas include household garbage, construction, cultivation, coastal tourism and industry (Zhang *et al.*, 2022). With population size and therefore plastic production and waste ever-growing it remains important to acknowledge and monitor those places inputting the most. Therefore, this region of interest is aligned with my mission objective of observing the plastic on and around the coast of Shanghai.

Payload

When choosing an imaging tool, it was important to find one that had a varied spectral range. This is because there are a multitude of different plastics in our region of interest and therefore different spectrums that need to be monitored. The main examples are shown in figure 2. This fact means this camera is suitable for this mission as it has a spectral range of 400nm-1000nm ('ORION12MP | satsearch', n.d.).

Plastic	Spectral range	Spectral band
Fiber	570-650nm	Visible light
Fragment	450-490nm	Visible light
Film	400nm	Visible light
Pellet	785-852nm	Infared radiation

Figure 2: shows the different plastics found in the region of interest, their spectral range and their spectral band **('ORION12MP | satsearch', n.d.).**

In addition, another reason for choosing this camera after thorough research is that it has 256 Gbytes on board memory, which would allow for a sufficient amount of data to be collected ('ORION12MP | satsearch', n.d.). This device can operate from a temperature of –20 degrees Celsius to 60 degrees Celsius, furthermore justifying this camera is the most suitable for this mission as it would be able to complete its orbits around Earth.

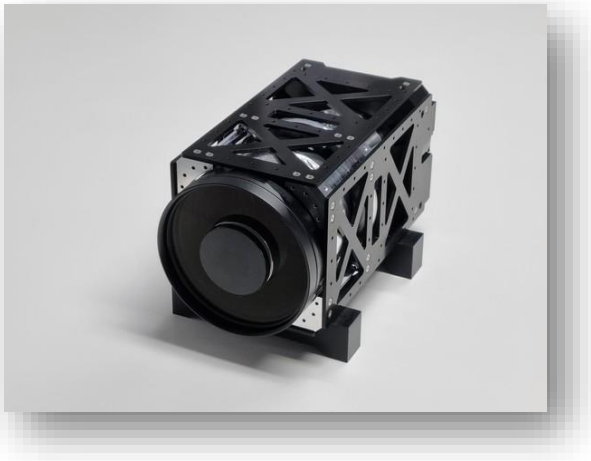


Figure 3: Orion 12MP camera (‘ORION12MP | satsearch’, n.d.)

Orion12MP	Value
Size	100mm*100mm*55mm
Mass	1.5kg
Band	Megapixel RGB
Swath	20km
Focal Length	0.55m

Table 1*: shows the different aspects of the camera* (‘ORION12MP | satsearch’, n.d.).

Calculating the mission parameters was done to help understand if what this mission is attempting to achieve can be done. Originally the swath was bigger but after vigorous debate it was changed. Having a swath of 20Km means that enough data can be collected for reproducible results and data over a wider range of time can be taken.

After researching the region of interest and understanding the types of plastic that are causing pollution: household items and construction to name a few, it was decided that a smaller GSD would allow the plastic to be viewed using this device. This is because it was found that items of plastic bottle shaped would be the most likely to be present. Further parameters like angular resolution, diameter and pixel size were also calculated and can be seen in *table 2*.

Mission parameters	Value
Swath	20Km
GSD	1m
Angular resolution	8.2 arcseconds
Diameter	22cm
Pixel size	11e-4

Table 2**:** shows the mission parameters

Payload

```
Altitude     = 500e3; % altitude (m)
Swath        = 20e3; % (m)
GSD          = 1; % (m)
A_r_deg      = atan(GSD/Altitude)% angular resolution in deg
A_r_rad      = deg2rad(A_r_deg)% angular resolution in rad
A_r_arc      = 206265*(A_r_rad) % (arcsecs)
lamda        = 852e-9; % Largest wavelength required
D            = 251643 * (lamda/A_r_arc); %camera diameter
Focal_length = 550-3; % (m)
pixelsize    = (GSD*Focal_length)/Altitude
FoV          = A_r_arc * Swath/GSD
Cam_d = 251643.3 * lamda/1
```

Figure 4: Payload calculations

Orbital Trade-Off

To ensure the satellite mission has optimal coverage time over the region of interest and the ground station, we have compared different iterations of orbital elements against the coverage time. Orbital elements such as the altitude, Inclination and RAAN will be adjusted thus providing the best orbital track. The main ground station that will be observed is the Parkes Observatory in Australia. Also, the satellite will use an on-board attitude determination sensor to point at the desired location.

Altitude vs Coverage

Altitude (Km)	No. Visits over Ground Station (Per 24 Hrs)	Total Time Coverage over Ground Station (Seconds)	No. Visits over Region of Interest (Per 24 Hrs)	Total Time Coverage over Region of Interest (Seconds)
400	7	3854.010	8	4321.447
500	7	4422.3454	8	5173.65
600	8	5223.843	8	5443.224

Table 3: *Altitude vs Coverage*

Inclination vs Coverage

Inclination (Degrees)	No. Visits over Ground Station (Per 24 Hrs)	Total Time Coverage over Ground Station (Seconds)	No. Visits over Region of Interest (Per 24 Hrs)	Total Time Coverage over Region of Interest (Seconds)
40	7	4272.373	8	5177.043
45	7	4422.34	8	5173.650
50	8	4543.179	8	4882.632

Table 4: Inclination vs Coverage

RAAN vs Coverage

RAAN (Degrees)	No. Visits over Ground Station (Per 24 Hrs)	Total Time Coverage over Ground Station (Seconds)	No. Visits over Region of Interest (Per 24 Hrs)	Total Time Coverage over Region of Interest (Seconds)
71	8	4550.080	8	5118.650
73	7	4422.344	8	5173.650
75	7	4485.753	8	5191.370

Table 5: RAAN vs Coverage

Final Orbital elements

Orbital Element	Value
Semi major axis	6971 Km
Eccentricity	0
Inclination	40 deg
Argument of perigee	115 deg
RAAN	75 deg
True Anomaly	0 deg

Table 6: Orbital Elements

No. Visits over Ground Station (Per 24 Hrs)	Total Time Coverage over Ground Station (Seconds)	No. Visits over Region of Interest (Per 24 Hrs)	Total Time Coverage over Region of Interest (Seconds)
7	4897.371	7	5161.770

Table 7: Orbital Elements vs Coverage

Above are the most optimal orbital elements for the mission and when they are combined, the total time of coverage for the ground station and region of interest is shown in Table 7.

Ground Station Location

To ensure the Parkes Observatory was the best option for this mission, the number of passes and time coverage was compared to two other ground stations, as shown in Table 8.

Ground Station	No. Of Passes	Time of Coverage (Seconds)
Parkes Observatory Australia	7	4897.371
Red1 – Redu Belgium	6	3188.244
ESA's Cebreros Deep Space Antenna	7	4540.809

Table 8: Ground stations vs Coverage

Parkes Observatory Australia had the most visits and the longest stay, based on the orbital parameters that were discussed in the section on orbital trade-off. Three ground stations were first chosen, and the ground station with the best coverage was chosen.

Field of View

The field of view analysis shows that the field of view distance varies with increasing altitude since it is correlated with both swath and altitude. The FOV distance increases along with its corresponding altitude.

Altitude (km)	Swath (km)	FOV (Km)
400	20	72.0
500	20	90.9
600	20	109.1

Table 9: FoV with change of altitude

Conclusion

In conclusion, this task involved selecting a region of interest and optimizing orbital elements to maximize coverage over Shanghai, a heavily polluted area. The mission design included choosing a suitable imaging tool, the Orion12MP camera, due to its spectral range and storage capacity. Detailed analysis of orbital parameters ensured optimal coverage time over the region and ground stations, particularly the Parkes Observatory. This approach ensures effective monitoring of plastic pollution in Shanghai's coastal and riverine environments.

References

Wang, M.H., He, Y. and Sen, B. (2019) Research and management of plastic pollution in coastal environments of China. *Environmental pollution (Barking, Essex : 1987)* [online]. 248 (1), pp. 898–905. Available from: https://www.ncbi.nlm.nih.gov/pubmed/30856505.

Luo, W., Su, L., Craig, N.J., Du, F., Wu, C. and Shi, H. (2019) Comparison of microplastic pollution in different water bodies from urban creeks to coastal waters. *Environmental Pollution* [online]. 246, pp. 174–182.

Zhang, F., Peng, G., Xu, P., Zhu, L., Li, C., Wei, N. and Li, D. (2022) Ecological risk assessment of marine microplastics using the analytic hierarchy process: A case study in the Yangtze River Estuary and adjacent marine areas. *Journal of Hazardous Materials* [online]. 425, p. 127960. [Accessed 28 July 2022].

Satellite Ground Station Map (vksdr.com) ---> used this to find austrailia ground station

'ORION12MP | satsearch' (no date) *satsearch.co* [online]. Available from: https://satsearch.co/products/infinity-avionics-orion12mp [Accessed 17 July 2024].